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May 28, 1975
SUPERSEDING
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July 9, 1964

FEDERAL SPECIFICATION

POLISHER-CLEANER-TARNISH PREVENTIVE, SILVER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers a liquid silver polisher-cleaner-tarnish preventive for use on sterling silver and silver plated articles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated And Solid, Sheet Stock (Container Grade), And Cut Shapes.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM):

ASTM D 92 - Flash and Fire Points by Cleveland Open Cup.
ASTM E 70 - pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Silver tarnish preventive and cleaner shall consist of a nonflammable liquid containing a finely divided abrasive suitable for polishing silver and containing a tarnish preventive material that remains on the silver after polishing.

3.2 Excluded material. Tarnish preventive and cleaner shall be free from cyanides, or any ingredients or solvents having a detrimental effect on silver or personnel, when tested as specified in 4.10.

3.3 Cleaning efficiency. The polish shall have a cleaning efficiency of not less than 85 percent when tested as specified in 4.4.1.

3.4 Tarnish resistance. When tested as specified in 4.5, the polish shall have a tarnish resistance of not less than 85 percent.

3.5 Abrasive number. The abrasive number shall not exceed 20 when tested as specified in 4.6.

3.6 Caking properties. When tested as specified in 4.7, the separated abrasive material shall be soft and easily redispersed by stirring or shaking.

3.7 Flammability. The flash point shall be not less than 140°F when tested as specified in 4.8.

3.8 pH. The pH shall be not less than 5.0 when tested as specified in 4.9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Certificates of compliance. Where certificates are submitted, the Government reserves the right to check-test such items to determine the validity of the certification.

4.3 Quality conformance inspections.

4.3.1 Quality conformance inspection sampling. Unless otherwise specified, sampling shall be in accordance with MIL-STD-105, at the inspection levels and acceptable

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quality levels (AQL's) indicated herein:

Paragraph	Inspection level	AQL
4.3.2	S-2	2.5
4.3.3	S-2	2.5
4.3.4	S-1	N/A

4.3.2 Inspection of preparation for delivery. An inspection shall be made to determine conformance to the requirements of section 5. The sample unit for this inspection shall be one filled shipping container, fully packed, marked, and ready for shipment.

Examine	Defect
Marking	Missing, incorrect, or illegible; not as specified.
Packing	Not as specified.
Packaging	Not as specified.
Contents	Net contents less than specified.
Workmanship	Weak or unsound seams; tears, cuts, splits, cracks, or holes; closure incomplete or inadequate; evidence of leakage or seepage from any part of the container; loose or inadequate strapping or stapling; bulged, distorted, or otherwise defective container.

4.3.3 Examination for closure and strapping of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure and strapping shall be in accordance with the appendix of that specification.

4.3.4 Inspection of the end item. An inspection shall be made to determine conformance to the material and workmanship requirements of section 3. The sample unit for this inspection shall be the contents of one filled unit container as specified in 5.1.1.

Examine	Defect
Material	Not a liquid.
Workmanship	Not uniform; not free from foreign matter.

4.3.5 Testing of the end item. The end item shall be tested for the applicable characteristics as indicated in table I for each lot presented for inspection. The sample unit shall be a 8- to 12-ounce bottle of polish. The sample size shall be five. Each sample shall be taken from a different shipping container of the lot. The test sample shall consist of a thirty-two-ounce composite of the five sample units. Duplicate determinations, unless otherwise specified, shall be performed on the composite. There shall be no evidence of failure to meet the specified requirements. Compliance with the requirement in 3.2 that the polish shall be free from any ingredients or solvents having detrimental affect on metals or personnel shall be satisfied by acceptance of a certificate of compliance from the supplier.

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TABLE I. Instructions for testing

Characteristic	Spec. par.	Reference test method	No. determinations per unit composite	Results reported as Pass or Fail 1/	Results reported as Numerically to nearest 2/
Cleaning efficiency	3.3	4.4.1	2	-	0.1 percent
Tarnish resistance	3.4	4.5	2	-	0.1 percent
Abrasive number	3.5	4.6	Avg. of 2	-	number
Caking properties	3.6	4.7	Avg. of 2	X	
Flammability	3.7	4.8	1	X	

1/ If failure is indicated, report the description of failure.
2/ Test reports shall include all values on which average results are based.

4.4 Test procedures.

4.4.1 Cleaning efficiency.

4.4.1.1 Materials and equipment.

(a) Smooth, rolled, sterling silver panels, 4 X 4 X 1/16 inches. The silver alloy shall be approximately 92 percent silver and 7 percent copper with a hardness of 30 to 50 on the Rockwell B Scale. The silver panels shall be suitably mounted on a firm base.
(b) A polishing machine having a 4-inch polishing head. The head shall have a cemented rubber pad (1/4 inch min) cemented to the head. A wrinkle free piece of billiard cloth shall be placed over the polishing head. The total weight of polishing head shall be 4.0 ± 0.1 pounds.
(c) A Hunter 60° Glossmeter or equivalent.
(d) A saturated solution of potassium sulfide.

4.4.1.2 Preparation of panels. Clean 2 silver panels using a jeweler's rouge slurry. The cleaned panels shall have a minimum of 77 percent specular reflectance at 60 degrees.

4.4.1.3 Procedure. Record the 60° specular gloss of the test panels. Tarnish the panels by immersion in the saturated potassium sulfide solution, maintained at 100° ± 2°F for 2 minutes. Rinse in distilled water and blot dry.

4.4.1.4 Apply 10 ml of the test polish to the polishing head and polish the panel for 114 strokes. Apply 5 ml more on polishing head and rotating the panel 90°, polish for 114 strokes. Repeat two more times. Rinse panels in distilled water and blot dry. Measure specular gloss of the panels.

$$\text{Cleaning efficiency, percent} = \frac{\text{reflectance of cleaned soiled panel} \times 100}{\text{reflectance of clean panel}}$$

4.5 Tarnish resistance test.

4.5.1 Preparation of panel. Take two clean silver panels (see 4.4.1.2) and apply the sample as provided in 4.4.1.4.

4.5.1.1 Test procedure. Measure the specular reflectance of the two polished panels and record. Place the panels in a dessicator in the presence of ammonium sulfide vapor for one hour. The ammonium sulfide shall be present as a water solution consisting of 100 ml of distilled water and 1 ml of a 1 percent solution of ammonium sulfide. Remove the panels at the end of test, rinse in distilled water, blot dry, and read the specular gloss of the two panels.

$$\text{Tarnish resistance, percent} = \frac{\text{Specular gloss of exposed panel} \times 100}{\text{Specular gloss of clean, polished panel}}$$

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4.6 Abrasive number. A piece of sulfur-free billiard cloth, 2-1/2 inches in diameter, shall be fastened securely in a smooth-bottom silver or brass container, 2-1/2 inches in diameter and 3/4 inch in depth, and covered to a depth of 1/4 inch with the polish. A sterling silver disk, 2 inches in diameter, buffed on one face and weighing approximately 60 g, shall be weighed and placed on the cloth. This disk shall be made of rolled silver, with a composition of approximately 92 percent silver and 7 percent copper, and a hardness on the Rockwell B Scale of 30 to 50. The disk shall be rotated at 55 to 60 revolutions per minute (rpm) under a weight of 4 pounds until 2,000 rotations have been completed. The disk shall then be removed, washed with soap and water, rinsed with distilled water, and wiped with a chamois or untreated paper or cloth. The weight loss in milligrams shall be taken as the abrasive number.

4.6.1 Sulfur-free billiard cloth. Prepare sulfur-free billiard cloth by washing the cloth in carbon disulfide for 5 to 10 minutes, followed by washing with benzene for 5 to 10 minutes. The washed billiard cloth shall be completely dried before use.

4.7 Caking of abrasive. Forty ml of the well-stirred polish shall be placed in a 75 ml cylindrical glass bottle, 1-1/2 inches in diameter, and centrifuged for 15 minutes at 1,500 rpm in a centrifuge having a 9 inch radius. The separated abrasive material in the bottle shall be examined with a glass rod.

4.8 Flammability. Flammability shall be determined in accordance with ASTM D 92.

4.9 pH. The pH shall be determined in accordance with ASTM E 70.

4.10 Cyanides. One drop of polish shall be added to approximately 0.10 g of a mixture of 90 percent anhydrous ferrous sulfate and 10 percent anhydrous ferric sulfate. The resulting paste shall be neutralized to litmus with dilute hydrochloric acid. A blue precipitate indicates the presence of cyanides.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A, B, or C. The polish shall be furnished in 8- to 12-ounce quantities packaged in a polyethylene commercial type squeeze bottle.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve 8- to 12-ounce bottles of polish shall be packed together in a box conforming to PPP-B-636, class 2. Fiberboard separators shall be provided with the fiberboard meeting the requirements of PPP-F-320. The box shall be closed and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. Twelve 8- to 12-ounce bottles of polish shall be packed together in a box conforming to PPP-B-636, class 1. Fiberboard separators shall be provided with the fiberboard meeting the requirements of PPP-F-320. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The compound, in the quantity specified (see 6.2), shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates, in containers complying with the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 General. Each container shall be marked with the directions for use.

5.3.2 Civil agencies. In addition to the markings required by 5.3.1 and any special markings required by the contract or order, each interior package and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.3 Military agencies. In addition to the markings required by 5.3.1 and any special markings required by the contract or order, each interior package and shipping container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The product is intended to be used for polishing and preventing tarnish on all sterling silver and silver-plated flatware and holloware.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Quantity required (see 5.1.1).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Documentation required.
(e) Special markings required (see 5.3.2 and 5.3.3).

MILITARY INTEREST:

The Department of Defense has waived coordination of this document until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
CPSC

PREPARING ACTIVITY: GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 30 cents each.

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